used in control system analysis.

Syllogistic Reasoning

In this mode of approximation reasoning, antecedents with fuzzy quantifiers are related to inference rules. This is expressed as—

x = S1 A's are B's

Y = S2 C's are D's

Z = S3 E's are F's

Here A, B, C, D, E, F are fuzzy predicates

- S1 and S2 are given quantifiers (fuzzy)
- S3 is the fuzzy quantifier which has to be decided.

Dispositional Reasoning

In this mode of approximation reasoning, the antecedents are dispositions that may contain the fuzzy quantifier "usually". The quantifier usually links together the dispositional and syllogistic reasoning; hence it plays an important role.

For eg., the propositional rule of inference in dispositional reasoning can be given as follows —

usually ((L, M) is R) $\Rightarrow$ usually (L is [R↓L])

Here [R↓L] is the projection of fuzzy relation R on L.